

Sales Trend and Inventory Optimization Analysis of Anand Pharma, Darbhanga

A Mid-Term Project Report | Primary Data Pathway (Jan–June 2026)

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1. Executive Summary and Title

Project Title: Sales Trend and Inventory Optimization Analysis of Anand Pharma, Darbhanga

Anand Pharma is a retail B2C pharmacy store established in 2018 and operated by Mr. Krishan Mohan Yadav in Donar, Darbhanga, Bihar (GSTIN: 10AEY3656A1Z4), procuring primarily from M/s. Sonu Pharma, Patna. The store encounters inventory challenges including seasonal demand fluctuations and intuition-based restocking decisions, resulting in stockouts of high-demand items, overstocking, and capital loss from expired medicines.

Primary sales, purchase, and stock data spanning 6 consecutive months (January to June 2026) were collected directly from physical vouchers and sales registers via remote digital transmission. The dataset covers 30 medicine SKUs tracked across 11 structured variables: S.No, Medicine Name, Category, MRP (₹), Purchase Price (₹), Opening Stock, Purchase Qty, Sales Qty, Closing Stock, Total Revenue (₹), and Expiry Date. Key descriptive statistics show mean revenue of ₹43,521.83 per SKU (median ₹41,108.50), total store revenue of ₹13,05,655, and an average gross margin of ~29.8%.

Quantitative techniques — ABC Analysis, FSN Classification, Pareto (80/20) Analysis, and 3-Month Moving Average Demand Forecasting — were executed. Preliminary findings reveal that 14 Category A medicines (46.7% items) generate 69.0% of revenue; 18 SKUs drive 80% cumulative revenue (Pareto cutoff); 5 Non-Moving SKUs carry high expiry risk; and May 2026 recorded peak revenue (₹2,43,998 — a 23.8% surge over April). Moving Average forecasts for July 2026 were generated for all SKUs to guide proactive procurement.

2. Proof of Data Originality

Primary data for 6 consecutive months (January 2026 to June 2026) was collected directly from M/s. Anand Pharma, Donar, Darbhanga, Bihar (GSTIN: 10AEY3656A1Z4; Proprietor: Mr. Krishan Mohan Yadav) through a remote digital pathway (scanned purchase tax invoices and daily sales registers shared via WhatsApp, cross-verified via telephonic calls).

- **Primary Sales, Purchase & Inventory Dataset (Jan–June 2026):** [\[Click Here\]](#)
- **Distributor Purchase Invoices & GST Details (M/s. Sonu Pharma):** [\[Click Here\]](#)
- **Pharmacy Firm Images & Servicescape Photos:** [\[Click Here\]](#)
- **Interaction Video (4-minute interaction with Proprietor in Hindi):** [\[Click Here\]](#)
- **Official Authorization Letterhead (Signed & Stamped):** [\[Click Here\]](#)
- **Academic Location & Remote Data Collection Clarification Note:** [\[Click Here\]](#)

2.5 Problem Statement and Business Objectives

Problem Statement 1: Minimizing Stockouts of High-Demand Medicines

Intuition-based purchasing causes stockouts of high-demand items (e.g., Ceroxim CV 500, Abzolid 600), leading to lost revenue. Objective: Apply ABC/Pareto classification to maintain 15-day safety buffer stocks for Category A items.

Problem Statement 2: Preventing Capital Loss from Unsold Expired Stock

Slow-moving SKUs risk expiring unsold, causing direct capital loss. Objective: Execute FSN classification to flag Non-Moving SKUs for credit returns to distributors 60 days before expiration.

Problem Statement 3: Optimizing Seasonal Procurement via Demand Forecasting

Monthly sales fluctuate seasonally (May 2026 revenue surged 23.8% to ₹2,43,998). Objective: Apply 3-Month Moving Average forecasting to adjust procurement 15-20% in advance of peak seasons.

3. Metadata and Descriptive Statistics

3.1 Metadata Definition and Dataset Structure

The dataset covers 6 monthly sheets (JAN2026 to JUN2026) across 30 SKUs. Compact metadata structure:

Sheet Name	Columns	Data Type	Units	Description
1. 'JAN2026'	Medicine Name	string	-	Brand / generic name of the medicine
2. 'FEB2026'	Category	string	-	Therapeutic category (Antibiotic, Gastric Care, etc.)
3. 'MAR2026'				
4. 'APR2026'	MRP	float	Rupees	Maximum retail selling price per strip/unit
5. 'MAY2026'				
6. 'JUN2026'	Purchase Price	float	Rupees	Wholesale procurement cost per strip/unit
	Opening Stock	integer	strips	Number of strips in stock at beginning of month
	Purchase Quantity	integer	strips	Number of strips purchased during the month
	Sale Quantity	integer	strips	Number of strips sold during the month
	Closing Stock	integer	strips	Number of strips in stock at end of month
	Total Revenue	float	Rupees	Gross revenue generated (Sale Quantity x MRP)
	Expiry Date	string	MM/YYYY	Expiration date printed on medicine packaging

Table 1: Metadata Definition — Compact Monthly Dataset Structure

3.2 Descriptive Statistics and Quantitative Summary

Statistical Metric	MRP (₹)	Purchase Price (₹)	6M Sales Qty (Strips)	6M Total Revenue (₹)
Sample Count (N)	30	30	30	30
Mean (Average)	₹192.30	₹134.87	223.40	₹43,521.83
Median	₹179.00	₹124.00	225.00	₹41,108.50
Standard Deviation	₹110.75	₹80.00	35.68	₹25,890.82
Minimum	₹45.00	₹28.00	152	₹7,605.00
Maximum	₹480.00	₹340.00	317	₹1,15,680.00
Range (Max - Min)	₹435.00	₹312.00	165	₹1,08,075.00

Table 2: Descriptive Statistics — 30 Medicine SKUs (Jan–June 2026)

Key Insights: 1. Mean revenue (₹43,521.83) > Median (₹41,108.50) confirms right-skewed revenue distribution where top SKUs drive majority sales. 2. Sales volume ranges from 152 to 317 strips (SD: 35.68), justifying FSN velocity segmentation. 3. Average gross profit margin per strip is ₹57.43 (~29.8%).

4. Detailed Explanation of Analysis Process / Method

4.1 Data Cleaning and Preprocessing

1. Product Standardization: Variant brand strings across monthly bills (e.g., "Ceroxim CV", "CEROXIM-500") were merged into unified standard SKU identifiers.
2. Wholesale Rate Imputation: Missing purchase rates were backfilled using master distributor tax invoices from M/s. Sonu Pharma, Patna.
3. Mathematical Balance Reconciliation: Every row verified via $\text{Closing Stock} = \text{Opening Stock} + \text{Purchase Qty} - \text{Sales Qty}$.
4. Expiry Date Standardization: Dates formatted to MM/YYYY objects for shelf-life countdown modeling.

4.2 Analytical Methodologies and Mathematical Abstractions

Method 1: ABC Analysis — Periodic Consumption Value $PCV = \text{Sales Qty} \times \text{Purchase Price}$. Ranked by cumulative revenue: Category A (top 70%), Category B (next 20%), Category C (bottom 10%). Linkage: Identifies priority buffer SKUs (PS1).

Method 2: FSN Analysis — 6-month sales velocity: Fast ($\text{Sales} \geq 250$), Slow ($190 \leq \text{Sales} < 250$), Non-Moving ($\text{Sales} < 190$). Linkage: Flags high-risk expiring inventory for returns (PS2).

Method 3: Pareto Analysis (80/20 Rule) — Identifies minimal SKU subset driving 80% revenue. Linkage: Focuses purchasing capital on vital few SKUs.

Method 4: 3-Month Moving Average Demand Forecast — $F_{(t+1)} = (S_t + S_{(t-1)} + S_{(t-2)}) / 3$. Linkage: Predicts seasonal demand shifts for advance ordering (PS3).

5. Results and Findings (Preliminary Insights)

5.1 ABC Classification & Pareto Analysis Findings

Total 6-month store revenue reached ₹13,05,655.00 across 30 SKUs. ABC breakdown: Category A (14 SKUs, 46.7% items) = ₹9,00,660.00 (69.0% revenue); Category B (8 SKUs) = ₹2,72,427.00 (20.9%); Category C (8 SKUs) = ₹1,32,568.00 (10.2%). Top 3 SKUs: Ceroxim CV 500 (₹1,15,680), Abzolid 600 (₹92,568), Chyawanprash 2KG (₹98,100).

Figure 2: ABC Classification Analysis
Anand Pharma (Jan-June 2026)

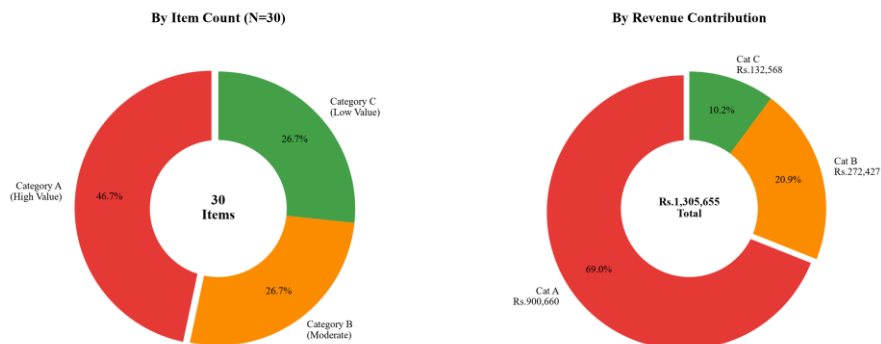


Figure 1: ABC Revenue & Item Distribution

Pareto 80/20 Cutoff: Exactly 18 out of 30 SKUs (60% items) generate 80% of cumulative store revenue, providing a clear cutoff for inventory capital allocation.

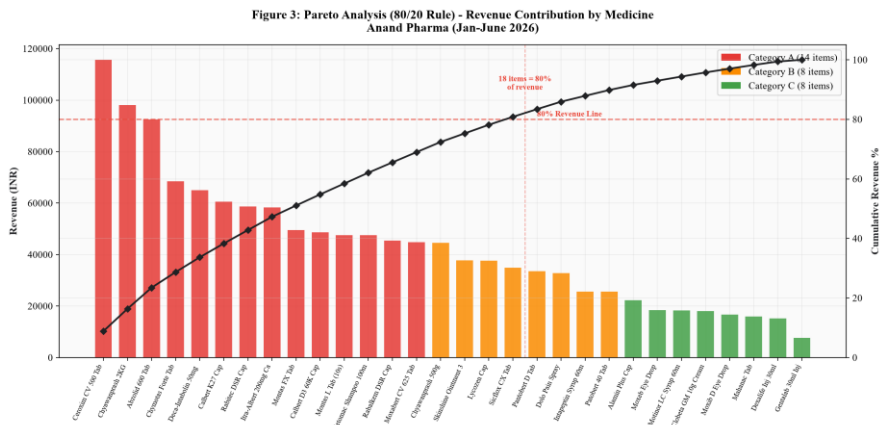


Figure 2: Pareto 80/20 Revenue Contribution

5.2 FSN Inventory Movement & Expiry Risk Findings

FSN Breakdown: Fast Moving F (5 SKUs, 16.7%), Slow Moving S (20 SKUs, 66.7%), Non-Moving N (5 SKUs, 16.7%). 5 Non-Moving SKUs carrying high expiry risk: Gentelab 30ml Inj (169 units), Dexalife Inj 30ml (208 units), Moxib D Eye Drop (152 units), Alamin Plus Cap (170 units), Dolo Pain Spray (168 units).

Figure 4: FSN (Fast-Slow-Non Moving) Inventory Classification
Anand Pharma (Jan-June 2026)

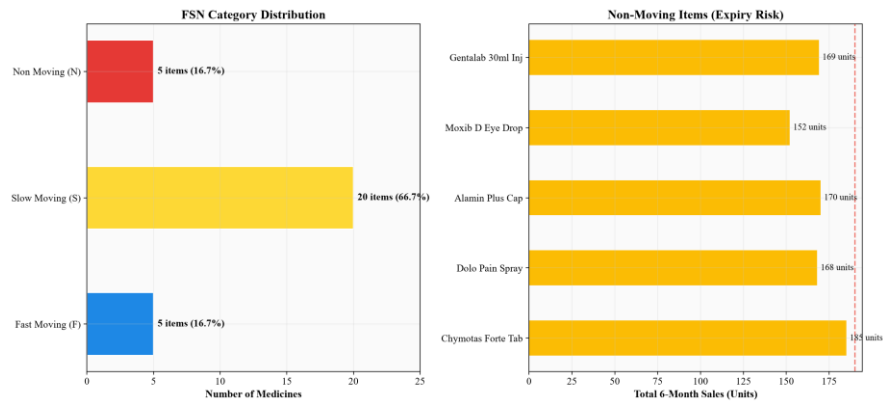


Figure 3: FSN Movement & Expiry Risk Items

ABC-FSN Matrix: Category A - Slow Moving items (10 SKUs) form the largest segment, requiring strict reorder point management to prevent stockouts.

Figure 9: ABC-FSN Cross-Classification Matrix
Anand Pharma (Jan-June 2026)

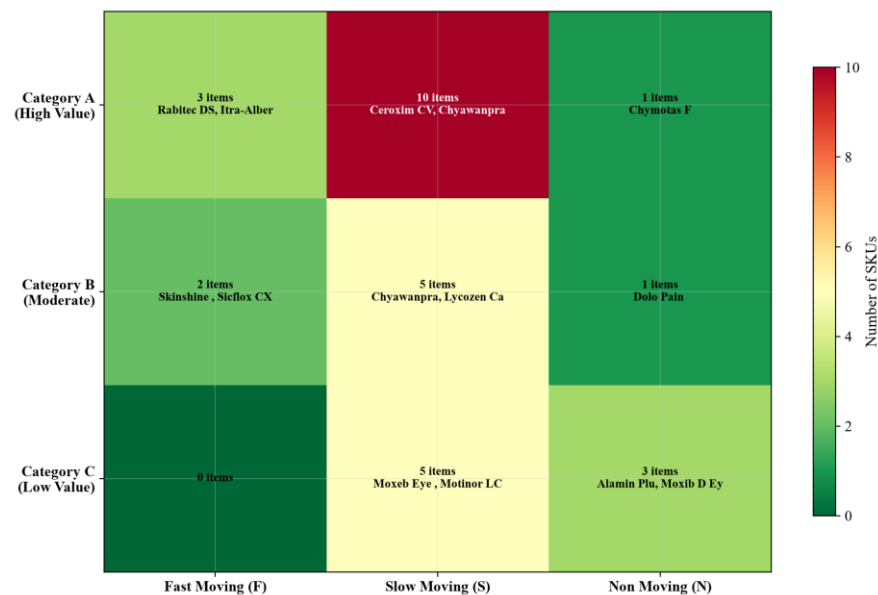


Figure 4: ABC-FSN Priority Matrix

5.3 Monthly Sales Trends & Seasonal Demand Fluctuations

Monthly Store Revenue: Jan (₹2,20,925), Feb (₹2,14,824), Mar (₹2,07,518), Apr (₹1,97,038 - lowest), May (₹2,43,998 - 23.8% pre-monsoon surge), Jun (₹2,21,352). Antibiotics (₹2,87,988) and Gastric Care (₹1,63,259) drive >34% of revenue.

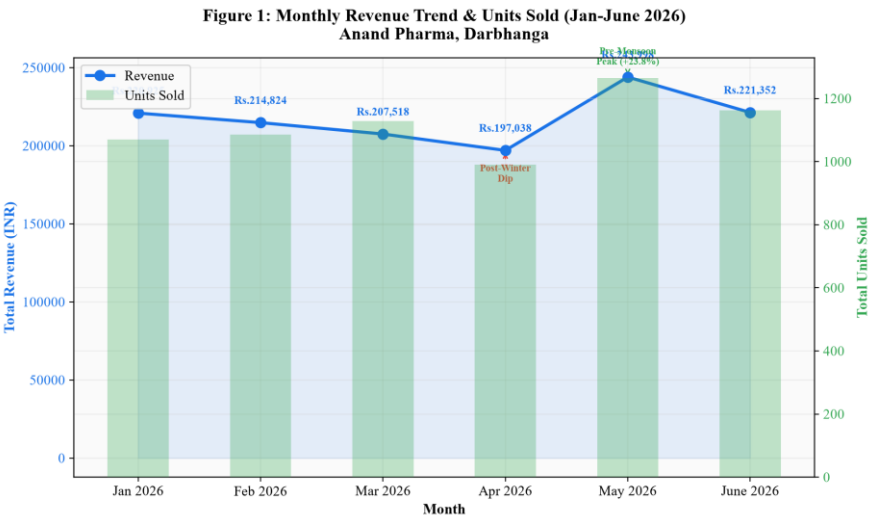


Figure 5: Monthly Revenue & Sales Volume Trend

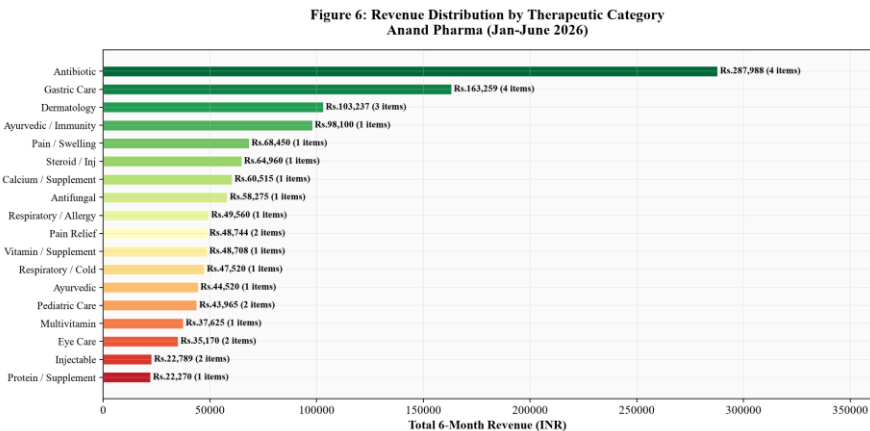


Figure 6: Revenue by Therapeutic Category

5.4 Moving Average Demand Forecast Results

July 2026 Demand Forecasts (3-Month SMA): Ceroxim CV 500 (45 strips), Abzolid 600 (37 strips), Chyawanprash 2KG (32 units - seasonal drop), Chymotas Forte (26 strips). SMA provides a conservative, lag-adjusted baseline for advance ordering.

Figure 5: 3-Month Moving Average Demand Forecast (Top 6 Category A Medicines)
Anand Pharma (Jan-June 2026)

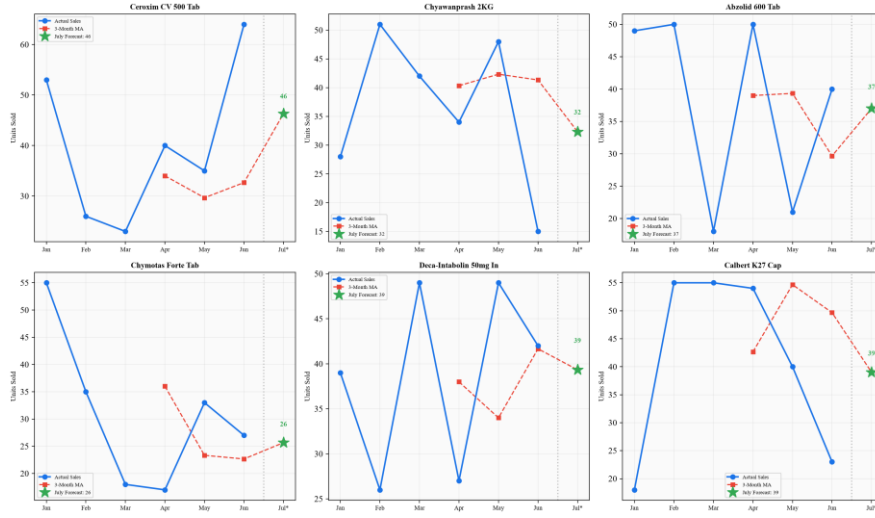


Figure 7: 3-Month Moving Average Demand Forecast (Top SKUs)

Figure 7: Sales Quantity vs Revenue (Bubble Size = MRP)
Anand Pharma (Jan-June 2026)

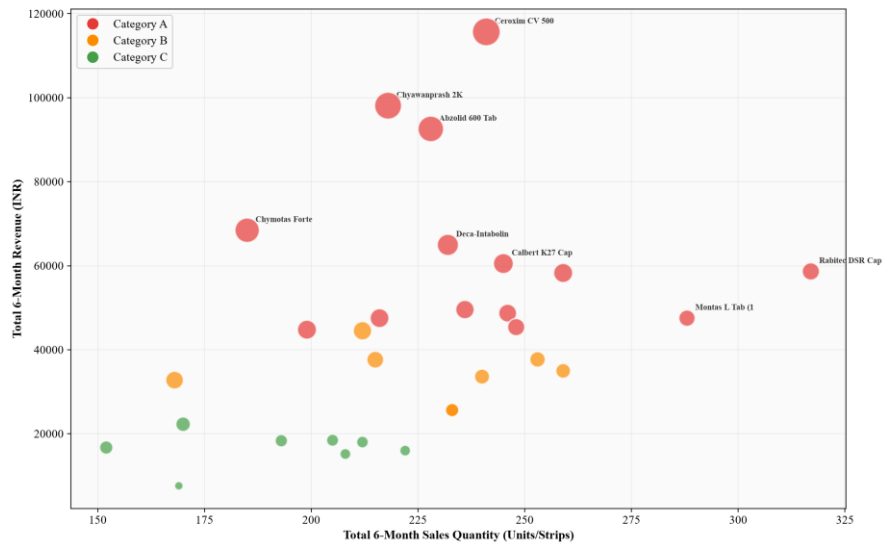


Figure 8: Quantity vs Revenue Bubble Chart (Size = MRP)

6. Interpretation of Results and Recommendations

1. Category A Safety Buffer Stocking (Addresses PS1): Maintain a 15-day safety buffer for Category A SKUs (Ceroxim CV, Abzolid) using $\text{Reorder Point} = \text{Avg Daily Sales} \times \text{Lead Time} + \text{Buffer}$.
2. Pre-Expiry Credit Returns (Addresses PS2): Implement a 60-day pre-expiry audit to return 5 Non-Moving SKUs (Gentalab Inj, Dexalife Inj) to distributor M/s. Sonu Pharma for credit notes.
3. Seasonal Advance Ordering (Addresses PS3): Increase Antibiotic and Gastric orders by 15-20% in April to capture the May-June pre-monsoon demand surge.
4. Digital Stock Audit Transition: Transition from manual registers to weekly Excel digital tracking for automated low-stock alerts.

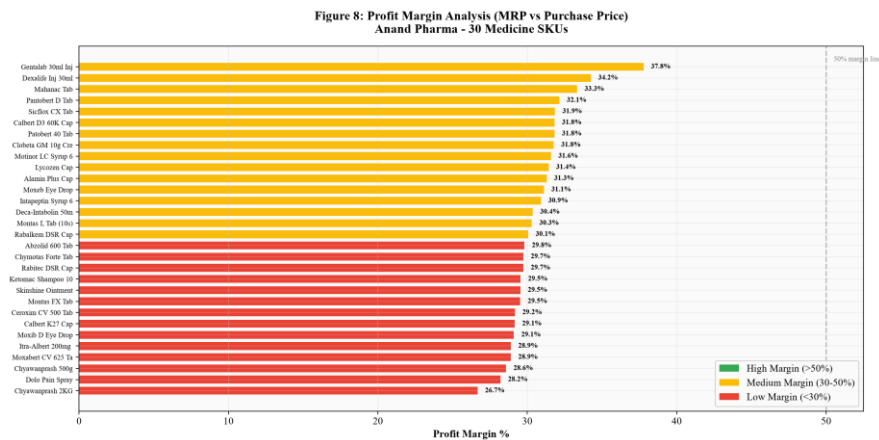


Figure 9: Profit Margin Analysis across 30 SKUs

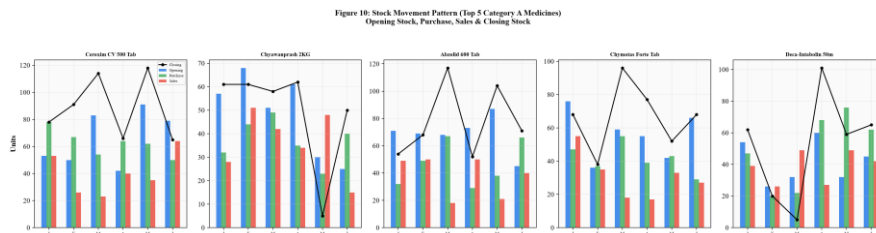


Figure 10: Stock Movement Pattern (Top 5 SKUs)